

Uber Data Analysis

Complete Project Documentation

Tool: Power BI Desktop

Category: Business Intelligence / Data Analytics Case Study

Deliverable: 3-page interactive Power BI dashboard (.pbix)

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1. Project Background

Ride-hailing companies like Uber operate on thin margins and razor-tight operational timing — a few minutes of driver misallocation or a poorly timed pricing decision can meaningfully affect revenue at scale. The data these platforms generate (trip timestamps, locations, distances, fares, vehicle types) holds the answers to when, where, and how to operate more efficiently — but only if it is structured and surfaced correctly.

This project was undertaken to simulate exactly that: take a raw Uber trip dataset and a business requirement document, and deliver a production-quality Power BI solution that a real operations or executive team could use to make decisions the same day it is opened.

The scope was deliberately structured around three distinct stakeholder needs rather than one flat report, because a single dashboard trying to serve executives, operations planners, and analysts simultaneously tends to serve none of them well.

2. Business Requirement Breakdown

The original requirement document specified the following, grouped into three functional areas:

Overview & Performance (→ Dashboard 1)

- Total Bookings, Total Booking Value, Average Booking Value
- Total Trip Distance, Average Trip Distance, Average Trip Time
- A dynamic measure selector for Bookings / Revenue / Distance
- Payment Type and Trip Type (Day/Night) breakdowns
- Vehicle Type performance grid with conditional formatting
- Daily booking trend
- Pickup/drop-off location intelligence, farthest trip, top 5 locations, preferred vehicle by location
- Supporting UX: data details bookmark, clear filters button, raw data export

Temporal Patterns (→ Dashboard 2)

- A global dynamic measure applied across all time-based visuals
- 10-minute interval demand curve
- Day-of-week comparison
- Hour × day heatmap

Granular Exploration (→ Dashboard 3)

- A detail-level grid of trip records
- Drill-through from any summary visual
- A bookmark to toggle between filtered and full data views

This grouping is what shaped the final information architecture — each dashboard answers a different kind of question rather than just a different subset of data.

3. Methodology

The project followed a standard BI development lifecycle:

```
Requirement Analysis → Data Understanding → Data Preparation →  
Data Modeling → Measure Development → Visualization → UX  
Enhancement →  
Testing → Documentation
```

Each stage is detailed in the sections below. This structured approach — rather than jumping straight into chart-building — is what allowed the dynamic measure selector and drill-through relationships to work correctly without rework.

4. Data Preparation (Power Query)

All transformation happened in Power Query before the data reached the model, following the principle of "clean once, use everywhere":

Step	Action	Purpose
1	Corrected data types (Date/Time, Decimal, Text)	Ensures DAX aggregations (SUM, AVERAGE) behave correctly
2	Decomposed Booking Timestamp into Date, Hour, Minute, Day Name	Powers the Time Analysis dashboard without repeated DAX calculations
3	Created a custom "10-Minute Interval" column by flooring the minute value	Feeds the interval area chart directly, avoiding a complex runtime measure
4	Added a custom sort key for Day Name (Monday = 1 ... Sunday = 7)	Ensures charts display in calendar order, not alphabetical
5	Filtered/flagged incomplete records (nulls in fare or distance)	Prevents skewed averages from bad data
6	Standardized column naming conventions	Improves model readability and downstream DAX clarity
7	Removed duplicate records (if any)	Ensures KPI accuracy

Update this table with the exact applied steps (visible in Power Query's "Applied Steps" pane) once finalized.

5. Data Modeling

5.1 Schema Design

The model follows a simplified star schema:

- Fact Table: Trips — one row per booking, containing measures (fare, distance, duration) and foreign keys (pickup location, drop-off location, vehicle type, payment type)
- Dimension Table — Date: standard calendar table for time intelligence

- Dimension Table — Location: unique list of all pickup/drop-off points
- Dimension Table — MeasureSelector: disconnected table (see 5.3)

5.2 The Role-Playing Dimension Problem

Both Pickup Location and Drop-off Location need to reference the same Location dimension table. Power BI restricts a table pair to a single active relationship, so:

1. Trips[PickupLocationID] → Location[LocationID] was set as the active relationship (used by default for all pickup-related visuals)
2. Trips[DropoffLocationID] → Location[LocationID] was created as an inactive relationship
3. A dedicated measure — e.g., Most Frequent Drop-off Point — wraps its calculation in CALCULATE(..., USERELATIONSHIP(Trips[DropoffLocationID], Location[LocationID])) to force Power BI to use the inactive relationship only for that specific calculation

This avoids duplicating the Location table (which would double storage and require separate maintenance) while still allowing both pickup and drop-off analysis from a single dimension.

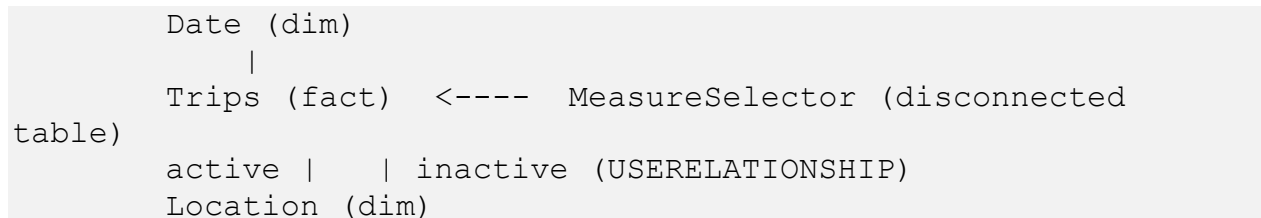
5.3 Disconnected Table for Dynamic Measures

To power the measure selector, a small standalone table was added to the model with no relationships to any other table:

MeasureName
Total Bookings
Total Booking Value
Total Trip Distance

Because it is disconnected, selecting a value in its slicer does not filter any data directly — instead, a DAX measure reads the selection via SELECTEDVALUE() and returns the result of a different calculation based on that selection. This is what allows one chart to represent three different KPIs without being rebuilt three times.

5.4 Relationship Diagram (Conceptual)



6. DAX Measures Reference

Measure Name	DAX Logic (representative)	Business Purpose
Total Bookings	COUNTROWS(Trips)	Core volume KPI

Total Booking Value	SUM(Trips[Fare])	Core revenue KPI
Average Booking Value	AVERAGE(Trips[Fare])	Revenue per trip
Total Trip Distance	SUM(Trips[Distance])	Aggregate distance
Average Trip Distance	AVERAGE(Trips[Distance])	Distance per trip
Average Trip Time	AVERAGE(Trips[Duration])	Duration per trip
Selected Measure	SWITCH(SELECTEDVALUE(MeasureSelector[MeasureName]), "Total Bookings", [Total Bookings], ...)	Powers the dynamic measure selector
Dynamic Title	SELECTEDVALUE(MeasureSelector[MeasureName]) & "Analysis"	Auto-updating chart titles
Most Frequent Pickup Point	CALCULATE(MAX(Location[LocationName]), TOPN(1, ...))	Location intelligence
Most Frequent Drop-off Point	CALCULATE(..., USERRELATIONSHIP(...))	Location intelligence via inactive relationship
Farthest Trip	MAX(Trips[Distance])	Outlier / edge-case detection
Top 5 Locations	RANKX(ALL(Location[LocationName]), [Total Bookings])	Demand concentration analysis

This table shows representative DAX patterns. Replace with the exact formulas (copied from Power BI's formula bar) for full accuracy.

7. Dashboard Build Process

Dashboard 1 — Overview Analysis

4. Placed six KPI cards for the core metrics at the top of the canvas
5. Added the MeasureSelector disconnected table as a slicer, wired to the Selected Measure DAX measure
6. Built Payment Type and Trip Type bar charts using Selected Measure as the value field
7. Bound each chart title to the Dynamic Title measure
8. Built the Vehicle Type matrix and applied conditional formatting to each KPI column

9. Built the "Total Bookings by Day" trend visual
10. Built location visuals: pickup point, drop-off point, farthest trip, top 5 locations, preferred vehicle by pickup location
11. Created a bookmark for the "Data Details" panel, linked to an info icon button
12. Added a blank button with the Clear All Slicers bookmark action
13. Added an export button (native "Export data" or a Power Automate-triggered flow)

Dashboard 2 — Time Analysis

14. Re-used the same MeasureSelector slicer for global consistency
15. Built the area chart on the 10-minute interval column from Power Query
16. Built the Day Name line chart, applying the custom sort-by column
17. Built the Hour × Day heatmap as a Matrix visual with conditional formatting applied

Dashboard 3 — Details Tab

18. Built a Table visual listing key trip fields (ID, date/time, locations, fare, distance, vehicle, payment type)
19. Enabled drill-through on this page, configured to accept context from Date, Location, and Vehicle Type
20. Added a "View Full Data" bookmark that clears drill-through filter context

8. Design Decisions & Trade-offs

Decision	Alternative Considered	Why This Approach Won
Disconnected table for dynamic measures	Building 3 separate charts per metric	Cuts visual count significantly, easier to maintain, cleaner canvas
USERRELATIONSHIP for drop-off analysis	Duplicating the Location dimension table	Avoids model bloat and keeps a single source of truth
Bookmarks for explanatory content	A static "About This Report" page	Keeps context available on demand without consuming canvas space
Matrix visual for the heatmap	A custom visual from AppSource	Keeps the report dependency-free and portable
Power Query-level date decomposition	Calculated DAX columns for Hour/Day Name	Better model performance — computed once at load time

9. Testing & Validation

- Cross-checked KPI totals against a manual pivot table / raw data sum to confirm DAX accuracy
- Verified the dynamic measure selector updates all bound visuals and titles correctly for each selectable metric

- Tested the inactive relationship measure by manually validating a sample of drop-off frequency counts against the raw dataset
- Validated drill-through by right-clicking multiple different data points and confirming the Details Tab filters correctly each time
- Tested edge cases — e.g., what the Farthest Trip and Top 5 Locations visuals show when slicers narrow the dataset to a small subset

10. Challenges & Solutions

Challenge	Solution
Power BI only allows one active relationship between two tables, but both pickup and drop-off locations needed the same dimension	Used an inactive relationship + USERELATIONSHIP() inside a targeted measure
Day Name sorting defaulted to alphabetical order, breaking the weekly trend narrative	Added a hidden numeric sort-order column in Power Query and applied "Sort by Column"
Needed one set of visuals to serve three different KPIs without triple the chart count	Implemented a disconnected MeasureSelector table with a SWITCH()-based measure
Users needed context on what each metric meant without cluttering the canvas	Used a bookmark-triggered info panel instead of permanent on-canvas text

11. Skills Demonstrated

- Requirement Analysis — translating a business requirement document into a structured technical scope
- Data Cleaning & Transformation — Power Query (M language) for shaping raw data into model-ready tables
- Data Modeling — star schema design, role-playing dimensions, disconnected tables
- DAX Development — dynamic measures, SWITCH, SELECTEDVALUE, USERELATIONSHIP, time intelligence
- Dashboard UX Design — bookmarks, buttons, drill-through, conditional formatting, dynamic titling
- Testing & Validation — cross-verification of calculated metrics against raw data
- Technical Documentation — clear, structured write-ups suitable for both technical and non-technical audiences

This document is intended to give a complete picture of not just what was built, but the reasoning behind each technical and design decision.